

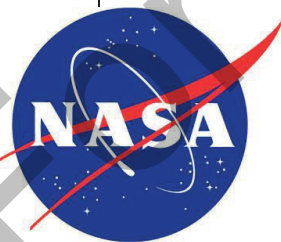
Balloon Program Office



Mobile Launch Vehicle Chassis Requirements

Effective Date

Date of Last Signature



**National Aeronautics and
Space Administration**

**Goddard Space Flight Center
Wallops Flight Facility
Wallops Island, Virginia**

Mobile Launch Vehicle Chassis Requirements

Signature/Approval Page

	Name	Signature	Date
Prepared by	Ed Udinski 820/Technology Manager, Balloon Program Office	Edward Udinski	Digitally signed by Edward Udinski Date: 2026.02.03 13:35:53 -05'00'
Reviewed	Leyland Young 511/NASA Engineering Representative	LEYLAND YOUNG	Digitally signed by LEYLAND YOUNG Date: 2026.02.03 13:56:18 -05'00'
Approved	Gabriel Garde 820/ Chief, Balloon Program Office	Gabriel Garde	Digitally signed by Gabriel Garde Date: 2026.02.03 14:08:58 -05'00'

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For reference only

1 INTRODUCTION

1.1 SCOPE

The purpose of this document is to establish and define design requirements for the New NASA Balloon Program Office (BPO) Mobile Launch Vehicles that will be used to launch balloons for domestic and foreign campaigns excluding Antarctica.

1.2 WORK DEFINITION

These vehicles will consist of two primary sections: the first consisting of the Power Plant and the second consisting of the Mobile Launch Vehicle Chassis. The Power Plant Section will consist primarily of the Engine, Transmission and Driver Cabin. The cabin will contain all the necessary controls to drive the vehicle as well as activate all the necessary hydraulic systems associated with such a vehicle. The Mobile Launch Vehicle Chassis Section will consist primarily of a Telescoping Lifting Boom, Electronic Observer Seat, and the Launch Director Deck.

This document will focus primarily on defining the requirements for the Mobile Launch Vehicle Chassis. This section, however, will need to accommodate a particular Power Plant Section that shall be commercially available in most parts of the world. The concept for these vehicles is to have the Power Plant Section interface with the Mobile Launch Vehicle Chassis during the campaign. Figure 1 below shows the sketch of one of the current Mobile Launch Vehicles (MLV) on site and identifies the primary sections being discussed.

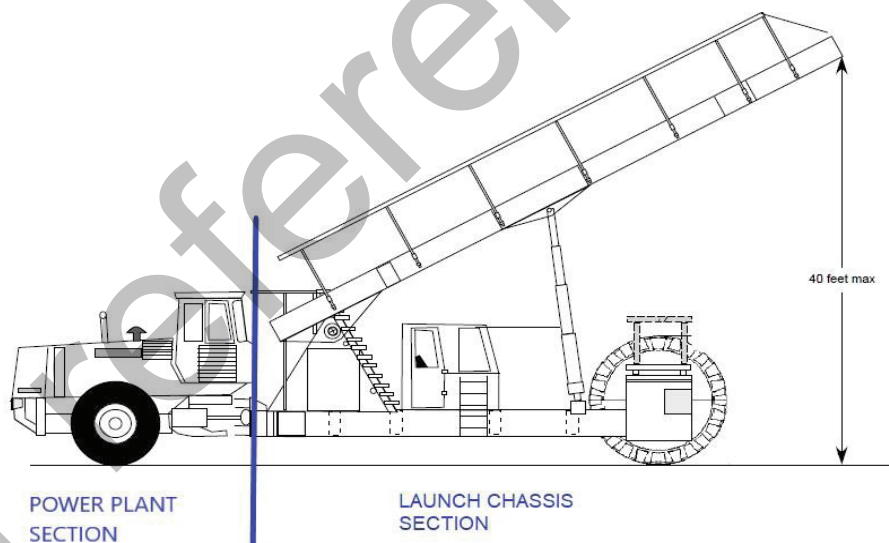


Figure 1- Mobile Launch Vehicle Sections

1.3 WORK ELEMENTS

1.3.1 Design

During the design phase the proper equipment configuration to yield the desired integration of the power plant and the Mobile Launch Vehicle Chassis, as well as component and

assembled unit operation. During this phase and prior to manufacturing, a review of the MLV chassis fabrication and construction drawings packages to include NASA representatives shall be completed. Deliverables for the review shall include:

1. Engineering drawings for the MLV Chassis, top level and for the various assemblies. Engineering drawings will include electronics copies of the MLV fabrication and construction drawings, as well as a step file of the MLV model that can be manipulated spatially to assist with payload integration.
2. An Acceptance Test Plan of the fully assembled MLV as a means of verifying the design, requirements, and capabilities of the system. The Acceptance Test Plan shall be inclusive of all the requirements levied.
3. A Post Acceptance Test Inspection plan of the fully assembled MLV to ensure its operational integrity.
4. Maintenance documentation for the MLV Chassis
5. Equipment cut sheets (equipment information sheet that provides and describes the specifications and/or features of a piece of equipment.) for major procured equipment components, applicable materials certifications, will be defined by the Contractor as part of the design reviews. Certifications will be provided to NASA BPO with the product quality data at completion.

1.3.2 Manufacturing

Prior to manufacturing beginning, a review of the major manufacturing milestones and critical material procurements will be held. Any design changes from the design review shall be reviewed prior to beginning manufacturing so as not to delay the start. During the manufacturing phase, the required materials and components will be acquired, fabricated as required and assembled into the final configuration. The unit shall be tested to procedures developed by the Contractor and approved by NASA. Maintenance and operational manuals, and mechanical Piping and Instrumentation Diagrams (P&ID's) and electrical schematics shall be completed and delivered with the initial unit. One copy of each document is to be delivered with each vehicle in both hard and soft copy. Updates to engineering drawings will be delivered to NASA BPO, as required.

2 RELATED DOCUMENTATIONS

Reference documents are to be selected and justified by the contractor. The lists included in this document are for reference only.

In the case of cancelled government documents, the latest issue will be used.

Listed documents that are not specifically cited in the text of this specification will be used for design guidance and in some cases ONLY sections of the cited documents will be found applicable.

2.1 REFERENCE DOCUMENTS GOVERNMENTAL

The following documents listed below shall form a reference list and may be found to be applicable to contractor design. Shall reference most current version.

- NASA-STD-5008 w/ Change 1: Protective Coating of Carbon Steel, Stainless, Steel, and Aluminum on Launch Structures, Facilities, and Ground Support Equipment.
- NASA-STD-5005: Standard for the design and fabrication of Ground Support Equipment
- NASA-STD-5020: Requirements for threaded fastening Systems in spaceflight hardware
- NASA-STD-8719.9 2: Lifting Standard
- NPR 7123.1: NASA Systems Engineering Processes and Requirements
- NPR 8715.1: NASA Safety and Health Programs
- GPR 1820.1: Hearing Conservation Program
- GPR 8715.8: Fall Protection Requirements for GSFC

2.2 REFERENCE DOCUMENTS NON-GOVERNMENTAL

The following documents listed below shall form a reference list and may be found to be applicable to contractor design.

- AISC 325: Steel Construction Manual
- ASCE-7: Minimum Design Loads and Associated Criteria for Buildings and Other Structures
- ISO 12944: Paints and Varnishes — Corrosion protection of steel structures by protective paint systems
- <https://buyquietroadmap.com/>
- AWS B1.11: Welding Inspection, Guide for the Visual Examination
- AWS 01.1: Structural Welding Code-Steel
- OM-300-200-03 SW.A: Launch Head Base Plate
- OSHA 29 CFR 1910.29: Fall Protection Systems and Falling Object Protection
- ANSI Z359: Fall Protection and Fall Restraint Standard
- OSHA 29 CFR 1910: General Industry Standards

3 REQUIREMENTS

3.1 GENERAL

The Mobile Launch Vehicle design will primarily satisfy TWO major requirements. The first requirement is a Power Plant Section that is readily available in the current locations that the NASA Balloon Program launches. The second requirement is satisfying the unique needs for this vehicle through the design of the Mobile Launch Vehicle Chassis Section.

It has been determined that one of the most readily available Power Plants is the one for the Caterpillar 966M Wheel Loader. Consequently, from this point forth, the Mobile Launch Vehicle Chassis Section requirements shall accommodate this specific power plant model.

Tailoring of the MLV requirements shall be formally documented as part of the contract and approved by the NASA Balloon Program Office. The request for tailoring shall include:

- a. The requirement(s) for which relief is being sought.
- b. A description of the request for relief (i.e., the nature of the proposed departure from the requirement).
- c. The reasons and justification for the relief, why it is not possible, or desirable, to comply with the requirement.
- d. The resulting change in risk.
- e. Confirmation the relief does not conflict with applicable Federal statutes or regulations, or with Agency policy or higher-level requirement.
- f. When and whether compliance will be achieved.
- g. Alternate actions to be taken for managing the risk.

3.2 SPECIFIC REQUIREMENTS

3.2.1 Dimensions/ Weight

1. The Mobile Launch Vehicle Chassis Section shall be designed in such a way that it does not allow the vehicle to tip over or otherwise allow the vehicle to be damaged if it shall experience the dynamic loads and speeds described in requirements 3.8.
2. It shall impart a maximum ground pressure of 80 psi to accommodate all launch locations.
3. The weight and dimensions for the Mobile Launch Vehicle Chassis Section shall be limited to allow loading and shipment via sea vessel and/or traditional highway transportation (special permits allowed)
4. The Mobile Launch Vehicle Chassis Section shall not be wider than 25 feet wide in its launch configuration.
5. The Launch Boom shall be 4 feet wide (Figure 2) and support a person weighing up to 350 lbs. walking on it. Flooring shall be see-through (current MLV uses expanded metal) and have compliant access such as steps with handrail.
6. All weights for major components shall be labeled accordingly and shall have appropriate structural lifting points and clearly marked Centers of Mass.
7. Any large or major components that may need to be removed for assembly or disassembly shall contain appropriate lift points or fork pockets as needed.

3.2.2 Clearances

1. The Mobile Launch Vehicle Chassis shall maintain a minimal ground clearance of 18 inches from the ground while carrying the maximum suspended load.
2. The Mobile Launch Vehicle Chassis Boom shall reach a height of at least 40 feet high but no more than 41 ft (Figure 2)
3. The projected Boom Height distance from the front end of the Mobile Launch Vehicle shall be no less than 12 feet (Figure 2).

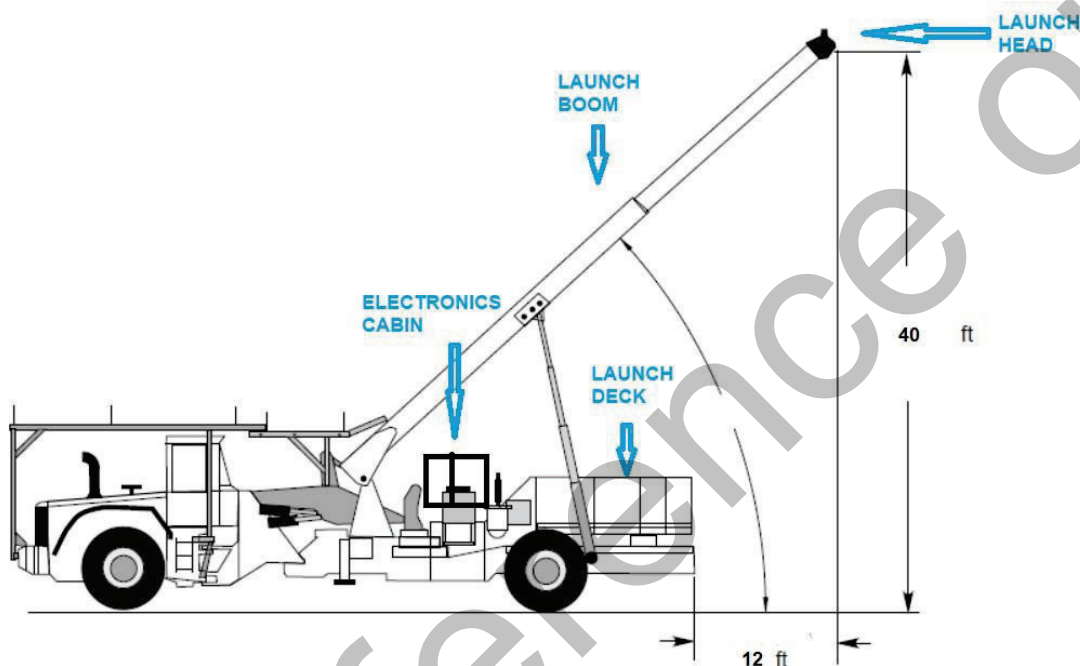


Figure 2- Mobile Launch Vehicle Hook Dimensions

3.2.3 Life Cycle

The Mobile Launch Vehicle Chassis shall have a life cycle of NO less than 30 years provided a reasonable maintenance plan is followed.

3.2.4 Specific Contractors Qualification

The Contractor shall provide evidence of the following minimum qualifications with their proposal to facilitate the proposal evaluation process:

3.2.4.1 Capability

Experience and/or background which demonstrates capability for accomplishing work of similar magnitude and complexity.

3.2.4.2 Description

Description of the Contractor's capabilities and expertise including but not limited to years of experience, similar work performed, personnel qualifications and backgrounds necessary for the manufacture of the Mobile Launch Vehicle Chassis.

3.2.4.3 Projects

A documented track record showing its ability to meet critical end dates and manufacturing schedules.

3.2.4.4 Resources

Adequate plant capacity and labor resources to ensure satisfactory completion of the proposed scope of work.

3.2.5 Quality Plan

To ensure that the requirements are met, the successful bidder shall have a documented quality assurance (QA) plan preferably certified to ISO 9000 and ISO 14000 standards which meet the following requirements:

1. Inspection procedures for receiving, in-process, and final inspection.
2. Calibration system that is traceable to the National Bureau of Standards (NBS) or have calibrations performed by a third-party calibration service with traceability to NBS.
3. Material identification and control procedure to include purchasing critical components directly from the manufacturer to prevent counterfeit parts.
4. Written test procedures / checklists

3.3 SCHEDULE REQUIREMENTS

The contractor shall specify and control design, manufacture, testing, delivery, and receipt of the units. The Contractor shall provide a schedule within two (2) weeks of contract award for both design and manufacturing phases. Refer to section 4.0 for additional schedule requirements details.

3.4 REQUEST FOR INFORMATION

The Contractor shall identify, to the BPO/NASA representative any recommended work, and associated costs to perform the work as indicated:

3.4.1 Exceptions

Specified work which does not meet the requirements of specifications referenced in Section 2.0 of this specification.

3.4.2 Hazards

Safety hazards discovered during manufacture, or within approved drawings or anticipated for future maintenance.

3.4.3 Discovery

Any perceived problem not designated above.

3.4.4 Other Scope

Any other work that the Contractor believes to be out of scope.

3.5 SAFETY REQUIREMENTS

1. The Mobile Launch Vehicle Chassis shall satisfy all Occupational Safety and Health Administration (OSHA) requirements for hand railings, steps, and any other applicable sections specifically on the fully extended launch boom and Mobile Launch Vehicle catwalks. The railing or system shall be designed in such a way that it does NOT present an entanglement source to the flight train (strands of wire rope).
2. Passive mechanical safety devices shall be utilized when applicable.
3. Any areas such as the payload deck area that requires access while a guardrail is down or not present (open edge) shall have an OSHA 29 CFR 1910 "Personal fall protection systems"/ANSI Z359 "Fall Protection Code" Series compliant restraint point for personnel to anchor to for fall restraint (at minimum).

3.6 DESIGN REQUIREMENTS

3.6.1 General

The intent of this specification is to provide design and performance requirements to develop and build a Mobile Launch Vehicle Chassis of adequate capability with low maintenance requirements and operating costs.

The Contractor shall make efforts to minimize the overall cost of the project, including adherence to commercial-build utilization where possible, while ensuring that performance, reliability, safety, and maintainability requirements are met. A brief conceptual design shall be included in the proposal.

3.6.2 Power Plate Selection

The key consideration in designing the Mobile Launch Vehicle Chassis is to understand the chosen commercial-off-the-shelf (COTS) power plant. This document requires the use of a Caterpillar 966M Wheel Loader selected based on a brief industry study of availability. The contractor has the responsibility to ensure that the power plant can integrate with the Mobile Launch Vehicle and meet all the requirements in this document. In the event that the Caterpillar 966M Wheel Loader cannot be integrated into the design, vendor shall notify BPO/NASA representative immediately and an in-depth justification will be required.

3.6.3 Design Products

In all cases, the design phase I drawing submittal phase will precede manufacturing or fabrication tasks. The engineering design drawings and schematics shall include but are not limited to the following:

- Mobile Launch Vehicle Chassis
- Specialized equipment and tooling required for operation and maintenance.
- Mobile Launch Vehicle Chassis systems schematics (braking, suspension, electrical)
- Hydraulic system schematic.
- Power systems and lighting schematic.
- Controls system schematic (alarms, emergency and safety features, process control, monitoring).

- Structural drawing package.
- Maintenance plan, electronic and hard copy; one set of each per vehicle delivered.
- Operation and assembly instructions, electronic and hard copy; one set of each per vehicle delivered.
- Power plant axle cart

3.6.4 Hydraulic System Design

The Mobile Launch Vehicle Chassis Hydraulic system will have to be independent from the Power plant (loader) and shall maintain the ability to operate when fully disconnected to run chassis jacks and boom. This feature is required for service and assembly/disassembly. Additionally, it is highly desired to have the controlling hydraulic system contained within a serviceable pallet for ease of eventual maintenance and repairs. The pallet shall include manual independent controls for:

1. Boom – In and Out motion
2. Boom – Up and Down motion
3. Chassis – Up and Down Lift Jacks

Furthermore, a wireless hydraulic control module shall be included to account for:

1. Boom – In and Out motion
2. Boom – Up and Down motion

3.6.4.1 Pistons

1. Pistons shall be sized to the appropriate operational factors of safety as justified by the contractor.
2. Pistons in boom structure shall incorporate appropriate, automatic safety measures to ensure safe operation and shutdown in the event of a hydraulic system failure.
3. Pistons shall not be housed in boom structure; such that disassembly of the boom is required for maintenance.
4. Positive blocks/lock-out points shall be incorporated in areas where equipment must be in a position that poses a crushing or pinch hazard to employees during maintenance or servicing.

3.6.4.2 Piping and Tubing

1. Hydraulic lines shall be sized to meet operational requirements laid out here as justified by the contractor.
2. Hydraulic lines shall be sized to the appropriate operational factors of safety as justified by the contractor.

3.6.4.3 Connections

1. All Mobile Launch Vehicle Chassis hydraulic components shall be selected to ensure direct integration with the power plant. Adaptors shall be avoided when reasonable.
2. Valves and caps shall be utilized to aid in assembly / disassembly to avoid undue spillage of hydraulic fluids.

3.6.4.4 Pipe Runs

1. Hydraulic lines shall be fabricated in lengths which are easily removed, easily transported, and configured to preclude unnecessary removal of hardware during maintenance. Metal lines shall be incorporated as much as possible.
2. Hydraulic line runs shall be mounted and secured to prevent rubbing between components or other tubing as well as preventing rubbing with the chassis.
3. Passive protection measures of hydraulic lines shall be utilized to prevent reasonably expected damage during normal operations, i.e. kick / cover plates.
4. Hydraulic lines shall not pass through or be run inside chassis structure.
5. Where feasible maintenance points, such as grease fittings, shall be designed to be accessible from the lowest level possible and not require personnel to be exposed to fall, crushing, or pinch hazards to access.

3.6.5 Operational Environment

3.6.5.1 Temperature

1. It is the contractor's responsibility to envelope and quantify conditions stated herein. Mobile Launch Vehicle Chassis shall be able to sustain operation in a cold, arid environment, such as a Sweden winter and will be bounded at -25 C.
2. Mobile Launch Vehicle Chassis shall be able to sustain operation in a hot environment, such as a Texas or Australian summer and will be bounded at 40 C.

3.6.5.2 Humidity/ Rain

The unit shall be capable of meeting all flow and temperature requirements in humidity ranging from 10% to 100%, which shall include rain rates up to 2"/hour.

3.6.5.3 Lightning

Chassis shall be appropriately grounded to provide a safe operational environment in the case of a lightning strike.

3.6.6 Protective Coating

Vendor shall refer to ISO 12944 for guidance on corrosion protection of the Mobile Launch Vehicle Chassis.

3.6.7 Fuel

All fuel used by and stored on the Mobile Launch Vehicle Chassis shall be independently plumbed to accommodate each subsystem requiring mechanical power. These subsystems are currently defined as a Gen Set and a separate engine to power the independent hydraulic system to support boom and chassis jacks

3.6.7.1 Fuel Tank

Due to the duration of flight line operations, two auxiliary fuel tanks supporting the Gen Set and hydraulic engine may be required to meet the operational needs.

1. Auxiliary Fuel tanks shall be located in such a way that it is easily plumbed to their respective systems
2. The minimum operation time of Gen Set shall be 16 hours.
3. The minimum operation time of hydraulic engine shall be 16 hours.

3.6.7.2 Piping Runs

- Fuels lines shall be fabricated in lengths which are easily removed, easily transported, and configured to preclude unnecessary removal of hardware during maintenance.
- Fuel line runs shall be mounted and secured to prevent rubbing between components or other tubing as well as preventing rubbing with the chassis.
- Passive protection measures of fuel lines shall be utilized to prevent reasonably expected damage during normal operations, i.e. kick / cover plates.
- Fuel lines shall not pass through or be run inside chassis structure.

3.6.8 Piping and Tubing

3.6.8.1 Materials

The Mobile Launch Vehicle Chassis piping, tubing, and components exposed to the environment shall be designed to withstand the operational environment defined in this document. Protective coatings shall be used only in those cases where it is not feasible to prevent corrosion by selection of materials of construction. It is the sole responsibility of the contractor to justify the material selection, but metal lines are preferred.

3.6.8.2 Supports

As a general rule, piping and tubing shall not be relied upon to support the weight of components and pressure gauges. Tubing on the equipment shall be restrained and supported using tube clamps, preferably the split-block type and spaced no more than 3 feet apart.

3.6.8.3 Fabrication

All piping and tubing shall be fabricated in lengths which are easily removed, easily transported, and configured to preclude unnecessary removal of hardware during maintenance.

3.6.8.4 Accessibility

1. Where flanged piping is used, the flange orientation shall be designed to minimize interferences with bolt installation and to provide adequate hand tool access. The piping system shall be designed such that individual components can be removed from the system (for maintenance or calibration) without dismantling or removal of adjacent components.
2. Ladders or platforms shall be incorporated in the design to aid in maintenance of equipment.

3.6.8.5 Protection

Passive protection measures of hydraulic lines shall be utilized to prevent reasonably expected damage during normal operations, i.e. kick / cover plates.

3.6.8.6 Filter and Gauges

Filters and gauges shall be located and oriented to allow for easy inspection.

3.6.9 Auxiliary Engine

1. The Mobile Launch Vehicle Chassis shall include a 12kw generator specified and adjustable to 120 V, 240 V and 50 Hz and 60 Hz.

2. The generator shall have the same operation requirements as the Mobile Launch Vehicle Chassis.
3. In accordance with, the “Buy Quiet by Design” (<https://buyquietroadmap.com>) the lowest dB producing generator that meets the performance requirements shall be considered for selection.

3.6.10 Electrical

3.6.10.1 General

Electrical system shall be designed for the operational environment defined herein.

3.6.10.2 Accessibility

The electrical system shall be designed such that individual components can be removed from the system (for maintenance or calibration) without dismantling or removal of adjacent components. Reusable disconnects shall be used whenever feasible.

3.6.10.3 Electrical Runs

1. Electrical runs shall be mounted and secured to prevent rubbing between components or other tubing as well as preventing rubbing with the chassis.
2. Passive protection measures of electrical runs shall be utilized to prevent reasonably expected damage during normal operations, i.e. kick / cover plates and conduit.
3. Electrical runs shall not pass through or be run inside chassis structure.

3.7 POWER PLANT INTERFACE

1. The Power Plant shall accommodate a wireless system to remotely control the launch boom hydraulic systems
2. The wireless system control box shall accommodate an emergency shutoff button for all hydraulic systems.
3. The wireless system control box shall be electrically powered by the Power Plant.
4. The wireless system control box shall include gauges for the independent hydraulic system to monitor: Engine oil pressure, temperature, charging and hydraulic pressure and temperature.

3.8 CHASSIS REQUIREMENTS

3.8.1 General

The Mobile Launch Vehicle Chassis consists of four main components: the chassis structure, the launch deck, the electronic technician jump seat, and the launch boom

3.8.2 Chassis Structure

The chassis structure shall be designed to withstand the static and dynamic loads, both specified and implied here in. It is the contractor’s responsibility to determine and justify appropriate factors of safety. The Mobile Launch Vehicle Chassis shall have a life cycle of NO less than 30 years provided the manufacturer’s maintenance plan is followed and operations fall within the designed boundaries. The vendor shall provide exceptions to the 30 year life cycle, based on the consumable or perishable nature of components. Maximum loading is predicted at 20 cycles annually at 60 minutes of operations per cycle. Operation cycles to include all worst-case dynamic loads as described by Section 3.8.2.1.2 of this document.

3.8.2.1 Loads

3.8.2.1.1 STATIC

1. The Mobile Launch Vehicle Chassis shall be designed in such way that it can sustain typical transport loads via highway or sea vessel. Additionally, the Mobile Launch Vehicle Chassis shall be able to be lifted and loaded into a traditional transport vessel without compromising its integrity.
2. The Mobile Launch Vehicle Chassis/Boom shall be able to lift a payload of at least 8,000 lbs. at its maximum extension and height.
3. The Launch Boom shall contain mechanical safety device(s) that allows it to maintain and secure the boom, while maximally loaded, during the lift / extension and at maximum elevation / extension. (Figure 3)
4. The Mobile Launch Vehicle (Chassis and Engine) shall be able to support the static loading conditions during inflation of up to 16,000 lbs of force while having a suspended payload weight of 8,000 lbs as shown in Figure 3. Loads shall be sustained at maximum height and extension of launch boom.

Static Load Case @ Max height and fully extended boom

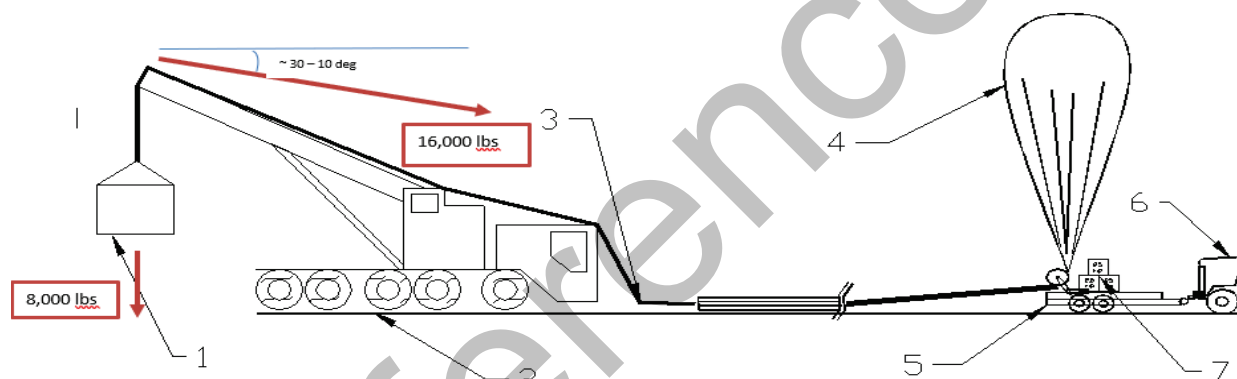


Figure 3- Static Load Case at Max Height and Fully Extended Boom

5. The Mobile Launch Vehicle Chassis/Boom shall be able support the Lift Equipment Configuration Certification (LECC) procedure and be to apply an upward static load of 16,000 lbs. at the boom head to a moored, dummy payload.

3.8.2.1.2 DYNAMIC

1. The Mobile Launch Vehicle shall be designed to a minimum factor of safety of 2 against deformation or yielding that impairs the function of the part and a minimum factor of safety of 3 against collapsing, buckling, exceeding the ultimate load, or failing to support the design load.
2. The Mobile Launch Vehicle shall be designed to sustain the following working loads shown in Figure 4 at the "16 knot" load case. Additionally, margins shall be used appropriately to prevent tipping of the Mobile Launch Vehicle while moving at a minimum speed range of 15 to 18 MPH.
3. Outriggers may be considered to help prevent tipping.

4. Dynamic analyses shall include; analysis of vehicle at top speed, both forward and reverse, with boom at worst-case loading and extension condition, tipping analysis, and loading under worst- case operational condition (e.g., at speed turning, hard braking, uneven road contains (inducing vibrational loading), etc.).
5. The following launch scenarios shall be used for worse-case boom loading during launch regarding horizontal (F_x), vertical (F_y), and lateral (F_z) force vectors along with an associated resultant force vector. Each scenario uses the following assumptions:

Balloon: A34.43-3H
 Gross Inflation: 14,406 lbf
 Drag Coefficients: Bubble = 2.3,
 Rope Section = 2.3 Aerodynamic Lift:
 Not Accounted
 For Net Lift: 8,510 lbf (gross lift minus balloon and flight train weight)

Scenario #1:

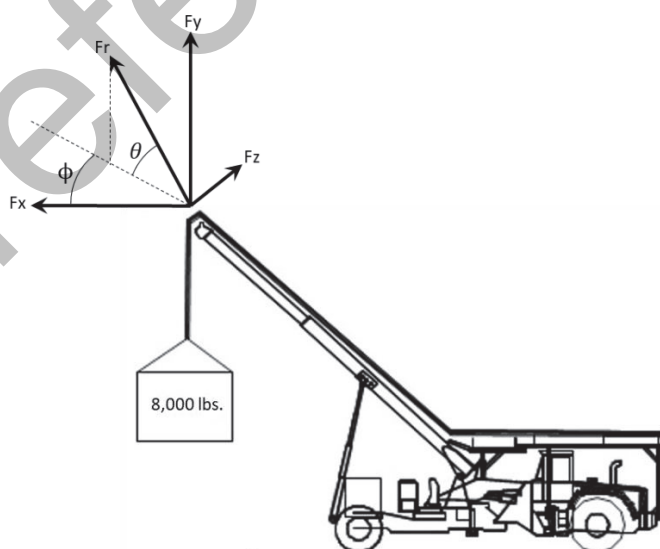
Balloon launch with winds of 16 kts and 0° crosswind (wind directly in launch direction).

Scenario #2:

Balloon launch with winds of 10 kts and 30° cross wind (wind in launch direction but offset by 30 degrees).

Scenario #3:

Balloon launch with winds of 10 kts and 90° cross wind (wind perpendicular to launch direction).



Scenario	Wind Speed [kts]	Fig. wind Dir [deg]	Forward Thr [lbf]	Launch Thr [lbf]	Dynamic Thr [lbf]	Fr [lbf]	θ [deg]	ϕ [deg]
#1	16	0	13909	8510	0	16306	31	0
#2	10	30	4705	8510	2717	10096	57	30
#3	10	90	0	8510	5433	10096	57	90

3.8.2.2 Speed

1. The Vehicle shall be capable of a minimum forward speed range of 15-18 MPH with the maximum suspended load at full boom extension.
2. The Vehicle shall be capable of a minimum Reverse speed of 10 MPH with the maximum suspended load at full boom extension.
3. The Vehicle shall be capable of transitioning from forward to reverse and vice versa with minimum mechanical lag and no electronic lag.
4. The transmission, overall acceleration will be as specified by the Caterpillar 966 M power train.
5. The Mobile Launch Vehicle will be steered through an articulated junction and shall have a minimum turning angle of 37 degrees.

3.8.3 Launch Deck

1. The launch deck shall be large enough to accommodate at least 6 people across the width of the deck and have an unobstructed view of the front end of the vehicle.
2. From the ground, the launch deck shall have a minimal deck height of 6 feet and allow an increase in height up to a minimum height of 12 feet.
3. Guard railing shall, per OSHA 29 CFR 1910.28 "Duty to have fall protection", encompass the perimeter of the launch deck with appropriate, self-closing gates at entrance points. Guardrails shall meet the design requirements defined in OSHA 29 CFR 1910.29 "Fall Protection Systems and falling object protection-criteria and practices".
4. The Front Deck shall contain a protective metal cage enclosure for both stationary occupancy and occupancy at top speed.
 - a. Protective cage shall be constructed out of 2 inch by 2 inch mild box steel with an 1/8 inch wall or equivalent.
 - b. Protective cage shall be designed to accommodate the height of a 98th percentile male.
 - c. Protective cage shall be designed with appropriate stand-off from the handrail to prevent pinch points.
 - d. Protective cage shall be designed in such a way to allow the boom to be completely lowered and stowed but provide full perimeter protection when the boom is raised.
 - e. Protective cage shall include a horizontal member on the payload facing side at the level of the topmost member of the handrail. Centerline of this member shall

fall within plus or minus the diameter of the handrail when measured vertically from the launch deck platform.

- f. Protective cage shall employ overhead protection capable of protecting occupants from falling debris while providing visibility to balloon and payload.
5. The Mobile Launch Vehicle chassis shall be able to provide a direct/hardline form of communication between the vehicle driver, the Electrical Technician Jump seat, and the Launch Deck located in the front of the vehicle (Figure 2)
6. NASA will take responsibility for providing and integrating MLV communication system.
7. The Whiskers System for control of the flight train during launch operations will consist of a minimum quantity sixteen (16) five-foot aluminum breakaway rods, divided between both sides of the launch boom, set at a 30-degree angle, spaced uniformly across the length of the Boom/Boom Extension. The system will consider all necessary safety and performance requirements in place during launch operations.

3.8.4 Electrical Technician Jump Seat

1. The Mobile Launch Vehicle chassis shall include an independent single person jump seat with seatbelts for the Electronic Technician to be seated during the launch attempt.
2. The jump seat shall be facing forward, located on the launch deck underneath the boom, and have a view of the forward operations as unobstructed as possible.
3. Every source of noise stemming from operation of the MLV chassis shall be considered and attenuated in such a way that the occupant of the electrical technician jump seat shall be subjected to a noise hazard no greater than 82 dB.
4. CSBF will take responsibility for providing and integrating MLV communication system.
5. The electrical technician jump seat shall incorporate a suspension method to minimize the chassis vibrations experienced by the occupant.
6. The electrical technician jump seat and enclosure shall be designed to accommodate a 98th percentile male.

3.8.5 Launch Boom

1. The Launch Boom shall be 4 feet wide and no wider than 4.5 feet.
2. The Mobile Launch Vehicle Chassis Boom shall reach a height of at least 40 feet high but no more than 41 ft high (Figure 2).
3. The projected Boom Height distance from the front end of the Mobile Launch Vehicle shall be no less than 12 feet (Figure 2).
4. The end of the boom shall support attachment for a NASA provided Launch Head Assembly as described in drawing OM-300-203 SW.A. The Launch Head shall remain centered along front end of the vehicle/boom. Furthermore, the boom shall accommodate a “through” hole pattern to secure the Launch Head to the Launch Boom.
5. The hydraulic cylinders supporting the extension of the boom shall be placed in such a way to reduce the possibility of oil leakage over the scientific instrument.
6. The Launch Boom shall contain the appropriate handrail system to allow work on top and shall extend the full length of the boom or to an agreed length. This safety feature shall meet OSHA 29 CFR 1910 “Walking-Working Surfaces” standards and not interfere with the launch process. Furthermore, it shall be designed to be stowed as to minimize any potential snag hazards to the flight train during launch.

7. The Mobile Launch Vehicle Chassis shall accommodate a suitable staircase ladder to allow access to the top of launch boom. This feature shall meet current necessary OSHA 29 CFR 1910 "Walking-Working Surfaces" standards.
8. Overhead cab 'Boom Walkway Extension' will consist of required safety measures including fall arrest points and safety railings. Dimensions will be greater than 108" wide in back and match the boom width in front (70" wide), with an expected load of 600lbs (personnel and equipment).

3.8.6 Working Lights

1. Appropriate lighting and distribution will be defined directly with vendor by NASA. These lights shall provide adequate visibility for nighttime operations at all points on interest; to include payload area, launch boom, launch deck, electrical technician jump seat, and generator at a minimum.
2. Working lights shall be adjustable and wired in zones.
3. Additional power circuits and ports shall be located at or near the generator and launch deck to allow for the addition of auxiliary lighting.

3.8.7 Assembly Plan

1. The Power Plant Section shall be able to mount directly to the Mobile Launch Vehicle Chassis Section through a similar method as shown in Figure 5.
2. The Mobile Launch Vehicle Chassis Section shall accommodate or support a tow/pull system with a class A Tractor Truck to allow the chassis movement without the Power Plant Section and shall accommodate a brake system to support such a movement, to include a parking brake system independent to the chassis. This tow/pull system shall be used to load and unload the Mobile Launch Vehicle Chassis to an appropriate trailer bed.
3. The Mobile Launch Vehicle Chassis Section shall be fully assembled using a mobile crane with a capacity of no more than 10 Ton and in the least amount of time possible, with a target assembly time of no more than 4 hours.
4. The Mobile Launch Vehicle Chassis Section shall allow for ease of assembly to the Power Plant through the use of a mechanical self-guided system similar to what has been accomplished by the Swedish Space Corporation in the "Hercules" vehicle.

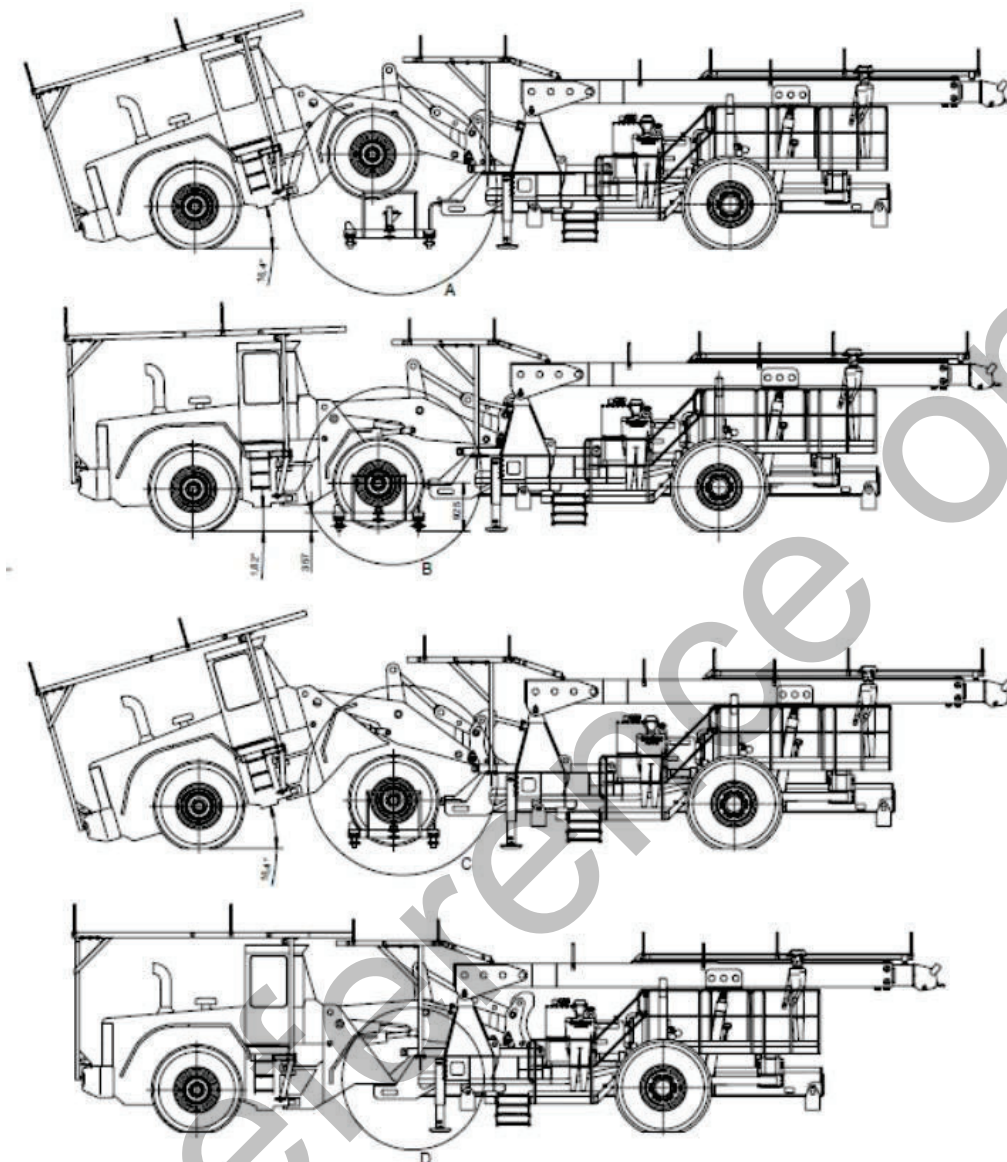


Figure 5-Mobile Launch Vehicle Section Separation

3.8.8 Fasteners/ Materials/ Welded Joints

1. All fasteners, and materials shall comply with AISC 326.
2. All welded joints shall comply with the manufacturing, inspection, and acceptance criteria of AWS D1.1M/D1.1M:2020 with specific inspection and acceptance criteria defined by PD per standard sounding rocket type launcher best practices.

3.8.9 Shipping/ Transportation

The Mobile Launch Vehicle Chassis section shall be designed in such a way that it can be legally (including special permits) transported via sea vessel and/or through major highways within the United States as well as countries such as New Zealand, Australia, Canada, and Sweden.

3.8.10 Testing

1. A proof test of the fully assembled Mobile Launch Vehicle Chassis shall be performed at 1.5 times the maximum designed load for both suspended and inflation loading.
2. Loaded maneuverability and stability testing shall be performed at top, operational speed. To include; forward, reverse and at minimum turning radius.
3. An Acceptance Test shall be performed at the manufacturing facility that is inclusive of all the requirements. NASA personnel shall be afforded the opportunity to attend.
4. A Post Acceptance Test Inspection shall be performed at the manufacturing facility and documented to ensure the operational integrity of the tested MLV chassis. The inspection shall be documented in a report and furnished to NASA for review. NASA personnel shall be afforded the opportunity to attend.

4 ENGINEERING REVIEW

An engineering review cycle is required as part of the MLV design process. This review cycle shall include a System Requirements Review (SRR), a Preliminary Design Review (PDR), and a Critical Design Review (CDR). The tables included in this section highlight the entrance and success criteria for each meeting. The following tables are adapted from Appendix G of NPR 7123.1C.

4.1 SYSTEM REQUIREMENTS REVIEW (SRR)

The SRR evaluates whether the functional and performance requirements defined for the MLV are responsive to the program's requirements and ensures the preliminary requirements will satisfy the MLV's operational and life cycle requirements. See Table 1

4.2 PRELIMINARY DESIGN REVIEW (PDR)

The PDR demonstrates that the preliminary design meets all system of interest requirements with acceptable risk and within the cost and schedule constraints and establishes the basis for proceeding with detailed design. See Table 2

4.3 CRITICAL DESIGN REVIEW (CDR)

The CDR demonstrates that the maturity of the design is appropriate to support proceeding with full-scale fabrication, assembly, integration, and test. The CDR determines that the technical effort is on track to complete the MLV development, meeting functional and performance requirements within the identified constraints at an acceptable risk. See Table 3

Table 1-System Requirements Require Entrance and Success Criteria

* In regard to interpretation of and proposed changes to this document.

System Requirement Review	
Entrance Criteria	Success Criteria
1. A preliminary SRR agenda, success criteria, and instructions to the review board have been agreed to by NASA. 2. Requirements for MLV Chassis are ready to be baselined after the review and preliminary review of flowed down requirements has been performed. 3. Other SRR work products have been made available to all cognizant participants *Updated concept definition. *Updated concept of operations. *Updated parent requirements. - Risk management plan ready to be baselined after review comments are incorporated. - Initial document tree or model structure. - Preliminary verification and validation method identified for each requirement. - Preliminary system safety analysis. - Preliminary Post Fabrication Acceptance Testing Plan. i. Interfaces with external systems are identified and preliminary definitions are ready to be baselined (e.g., Interface Control Documents). j. Cost and schedule data with ranges and a basis of the estimates.	1. The functional and performance requirements defined for the system are responsive to the stakeholder needs and parent requirements, reflect the systems intended operational use, and represent capabilities likely to be achieved within the scope of the project. 2. The maturity of the requirements definition and associated plans is sufficient to begin MLV design. 3. The project utilizes a sound process for the allocation and control of requirements throughout all levels, and a plan has been defined to complete the requirements definition at lower levels within schedule constraints. 4. System Interfaces with external entities (power plant) and between major internal elements have been identified. 5. Preliminary approaches have been determined for how requirements will be verified and validated. 6. Major risks have been identified and technically assessed, and viable mitigation strategies have been defined. 7. The program/project has demonstrated compliance with applicable NASA and implementing Center requirements, standards, processes, and procedures. 8. TBD and TBR items are clearly identified with acceptable plans and schedule for their disposition. 9. Proposed tailoring is appropriate and consistent with applicable Agency and Center guidance. 10. Lessons Learned from other projects and programs have been identified and addressed. 11. Single Point Failure/Fault Tolerance philosophy is reflected in requirements.

Table 2-Preliminary Design Review Entrance and Success Criteria

Preliminary Design Review	
Entrance Criteria	Success Criteria
1. The Project has successfully completed the previous planned life-cycle reviews, and all Action Items from the SRR have been addressed and resolved, or a timely closure plan exists for those remaining open. 2. A preliminary PDR agenda, success criteria, and instructions to review board have been agreed to by NASA. 3. All lower level peer reviews have been successfully conducted, and Action Items have been addressed. 4. The following primary products are ready for review: a. A preliminary design that can be shown to meet all technical requirements and performance measures. 5. Other PDR technical work products have been made available to the cognizant participants prior to the review: a. Subsystem design specifications with supporting trade-off analyses and data, as required, that are ready to be baselined after review comments are incorporated. b. Updated Risk Management Plan. c. Updated cost and schedule data d. Updated Post Fabrication Acceptance Testing Plan. e. Applicable design standards that have been	1. The top-level requirements are agreed upon, finalized, stated clearly, and consistent with the preliminary design. 2. The flow down of verifiable requirements is complete and proper, or, if not, an adequate plan exists for timely resolution of open items. Requirements are traceable to parent technical requirements and to mission goals and objectives. 3. The program/project cost and schedule are credible and within program/project constraints. 4. The preliminary design is expected to meet the requirements at an acceptable level of risk. 5. Definition of the system interfaces (both external entities and between internal elements) is consistent with the overall technical maturity, associated risks have been identified, and represents an acceptable level of risk. 6. The project risks are understood and have been credibly assessed, and plans, a process, and resources exist to effectively manage them and indicate that the program/project safety/reliability residual risks will be at an acceptable level. 7. Safety (e.g., safety, reliability, maintainability, quality controls, and quality verifications) have been adequately addressed in preliminary designs. 8. Adequate technical and programmatic margins and resources exist to complete the development within budget, schedule, and known risks. 9. The operational concept is technically sound, includes (where appropriate) safety systems, and includes the flow down of requirements for its execution. 10. The project has demonstrated compliance with applicable NASA and implementing Center requirements, standards, processes, and procedures. 11. TBD and TBR items are clearly identified with acceptable plans and schedule for their disposition. 12. Preliminary analysis of the primary subsystems has been completed and

identified and incorporated. f. Preliminary engineering drawing tree. g. Interface control documents that are ready to be baselined after review comments are incorporated. h. Verification/validation plan that is ready to be baselined after review comments are incorporated. i. List of potential single point failures.	summarized, highlighting performance and design margin challenges. 13. Appropriate modeling and analytical results are available and have been considered in the design. 14. Heritage designs have been suitably assessed for applicability and appropriateness. 15. Manufacturability has been adequately included in design.
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Table 3-Critical Design Review Entrance and Success Criteria

Entrance Criteria	Critical Design Review	Success Criteria
1. The project has successfully completed the previous planned life-cycle reviews, and all Action Items have been addressed and resolved or a timely closure plan exists for those remaining open. 2. A preliminary CDR agenda, success criteria, and instructions to the review board have been agreed to by NASA. 3. All lower level peer reviews have been successfully conducted and Action Items have been addressed. 4. Programmatic products are ready for review at the maturity levels required for identify a MLV fabricator. 5. A baselined detailed design that can be shown to meet all technical requirements and performance measures. 6. Other CDR technical work products (as applicable) for hardware, software, and safety system elements have been made available to the cognizant participants prior to the review: a. Product build-to specifications along with supporting trade-off analyses and data that are ready to be baselined after review comments are incorporated. b. Fabrication, assembly, integration, and test plans and procedures are being developed and are ready to be baselined after review comments are incorporated. c. Technical data package (e.g., integrated	1. The detailed design is expected to meet the requirements with adequate margins. 2. Interface control documents are sufficiently mature to proceed with fabrication, assembly, integration, and test, and plans are in place to manage any open items. 3. The program/project cost and schedule estimates are credible and within program/project constraints. 4. High confidence exists in the product baseline, and adequate documentation exists or will exist in a timely manner to allow proceeding with fabrication, assembly, integration, and test. 5. The product verification and product validation requirements and plans are complete. 6. The testing approach is comprehensive, and the planning for system assembly, integration, test, and mission operations is sufficient to progress into the next phase. 7. Adequate technical and programmatic margins and resources exist to complete the development within budget, schedule, and known risks. 8. Risks to safety and mission success are understood and credibly assessed and plans and resources exist to effectively manage them and indicate that the program/project safety/reliability residual risks will be at an acceptable level. 9. Safety and mission assurance (e.g., safety, reliability, maintainability, quality controls, and QA) have been adequately addressed in system and operational designs. 10. The program/project has demonstrated compliance with applicable NASA and implementing Center requirements, standards, processes, and procedures. 11. TBD and TBR items are clearly identified with acceptable plans and schedule for their disposition. 12. Material properties tests are completed as necessary along with analyses of loads, stress, finite element analyses, and other analyses. 13. The operational concept has matured, is at a CDR level of detail, and has been	

schematics, interface control documents, engineering analyses, and specifications). d. Status of technical performance related to margins and resolution of the previous review discrepancies addressing effectiveness of technical achievement and communicating the overall risk to the project. e. Defined operational limits and constraints. f. Updated technical resource utilization estimates and margins. g. Acceptance plans that are ready to be baselined after review comments are incorporated. h. Updated verification plan. i. Updated validation plan. j. Updated risk assessment and mitigation. k. Updated Life-Cycle Costs. l. Subsystem-level and preliminary operations safety analyses that are ready to be baselined after review comments are incorporated. m. Systems and subsystem certification plans and requirements (as needed) that are ready to be baselined after review comments are incorporated. n. List of all single point failures and their effects as well as rationale for acceptance.	considered in test planning. 14. Manufacturability has been adequately included in design. 15. Procurement and supply chain risk management execution is complementary with the technical development schedule.
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Appendix A Acronyms

BPO	Balloon Program Office
CDR	Critical Design Review
COTS	Commercial-Off-the-Shelf
GSFC	Goddard Space Flight Center
MLV	Mobile Launch Vehicles
OSHA	Occupational Safety and Health Administration
NBS	National Bureau of Standards
PDR	Preliminary Design Review
P&IDs	Piping and Instrumentation Diagrams
QA	Quality Assurance
SRR	System Requirements review
WFF	Wallops Flight Facility